Lexical Diversity Measurement

Across Genres

A Comparative NLP Study Using the Brown Corpus

Student
Ashish Singh

Subject
NLP

Dataset
Brown Corpus · 1M+ tokens

Tools
Python · NLTK · Pandas · Matplotlib

v4 · Enhanced Edition

github.com/4uashish2628/project-nlp

What is Lexical Diversity?

Definition

Lexical diversity measures the variety of vocabulary used in a text — the ratio of unique word types to total tokens.

High diversity → rich, varied vocabulary
Low diversity → repetitive, formulaic text

Key Applications

Authorship profiling · Language acquisition
Corpus linguistics · Discourse analysis
Text difficulty scoring · NLP benchmarking

The TTR Problem

TTR = Types ÷ Tokens

Humor (500 tokens) — **TTR = 0.71**
← *looks diverse*

Government (8k tokens) — **TTR = 0.31**
← *unfairly penalized*

TTR ∝ −0.764 correlation with text length
→ Longer texts are always unfairly penalized

Solution: Length-robust metrics → MATTR, HDD, MTLD

Project Objectives

Corpus Coverage

Measure lexical diversity across all 15 Brown Corpus genres (1,005,119 tokens)

Multi-Metric Comparison

Compare 8 metrics: TTR, Root TTR, Log TTR, Maas, MSTTR, MATTR, HDD, MTLD

Metric Reliability

Identify which metrics remain reliable despite unequal text lengths across genres

Innovation Experiment

Repeated equal-length sampling to empirically measure metric stability (CV analysis)

Composite Ranking

Z-score normalization across metrics → single composite diversity ranking for all 15 genres

Literature Survey

Francis & Kucera (1961/1979)
Established the multi-genre English corpus benchmark. Genre sizes are intentionally unequal — making robust metrics essential for fair comparison.

Foundation

Covington & McFall (2010)
Introduced MATTR (Moving-Average TTR). Reduces length sensitivity significantly; optimal window size requires empirical validation per corpus.

MATTR

McCarthy & Jarvis (2010)
Validated MTLD, vocd-D, and HD-D. MTLD is length-independent; shown more discriminative than raw TTR across genre-diverse corpora.

MTLD / HDD

Fergadiotis et al. (2015)
Confirmed MTLD and MATTR robustness under deliberate length variation in clinical language data — supporting cross-genre use.

Validation

Jarvis (2013)
Argued that lexical diversity is multidimensional; no single metric captures all facets — motivating our 8-metric comparative approach.

Framework

Consensus: Raw TTR is inadequate. MATTR, HDD, and MTLD are preferred for length-robust cross-genre comparison.

Implementation: Pipeline & Architecture

01 Data Acquisition
Brown Corpus via NLTK · 15 genres · 1,005,119 total tokens

02 Preprocessing
Lowercase normalization · Punctuation & non-word removal

03 Metric Computation
8 metrics per genre · TTR / Root / Log / Maas / MSTTR / MATTR / HDD / MTLD

04 Statistical Analysis
Z-score ranking · Spearman/Pearson correlation matrix · Stability CV

05 Export & Visualise
6 CSVs + 4 PNG plots + TXT summary written to results/

Generated Output Files

- genre_lexical_diversity.csv
- metric_rankings.csv
- stability_summary.csv
- metric_correlations.csv

Key Charts

- mtld_by_genre.png
- mattr_hdd_comparison.png
- metric_stability_comparison.png
- composite_diversity_score.png

```
# Core pipeline  (scripts/analyze_brown_corpus.py)

genres = brown.categories()                          # 15 genres
tokens = [w.lower() for w in brown.words(cats=g)    # normalize
          if w.isalpha()]
scores = compute_all_metrics(tokens)                 # 8 metrics
```

8 Metrics Explained: From Baseline to Robust

Metric	Tier	Formula	Note	Mean CV
TTR	Baseline	`Types ÷ Tokens`	Simple but length-sensitive	0.046
Root TTR	Classic	`Types ÷ √Tokens`	Partially corrects for length	0.038
Log TTR	Classic	`log(Types) ÷ log(Tokens)`	Log-scaled length correction	0.035
Maas	Classic	$(\log T - \log V)^2 \div \log T^2$	Lower = more diverse; inverted scale	0.033
MSTTR	Improved	`Mean TTR over segments`	Equal fixed-length segment averaging	0.019
MATTR	Robust	`Sliding window TTR mean`	Stable, CV = 0.0093	0.009
HDD	Robust	`Hypergeometric sampling`	Most stable, CV = 0.0071	0.007
MTLD	Robust	`Seq. length before drop`	Length-independent, higher = more diverse	0.068

Analysis & Key Findings

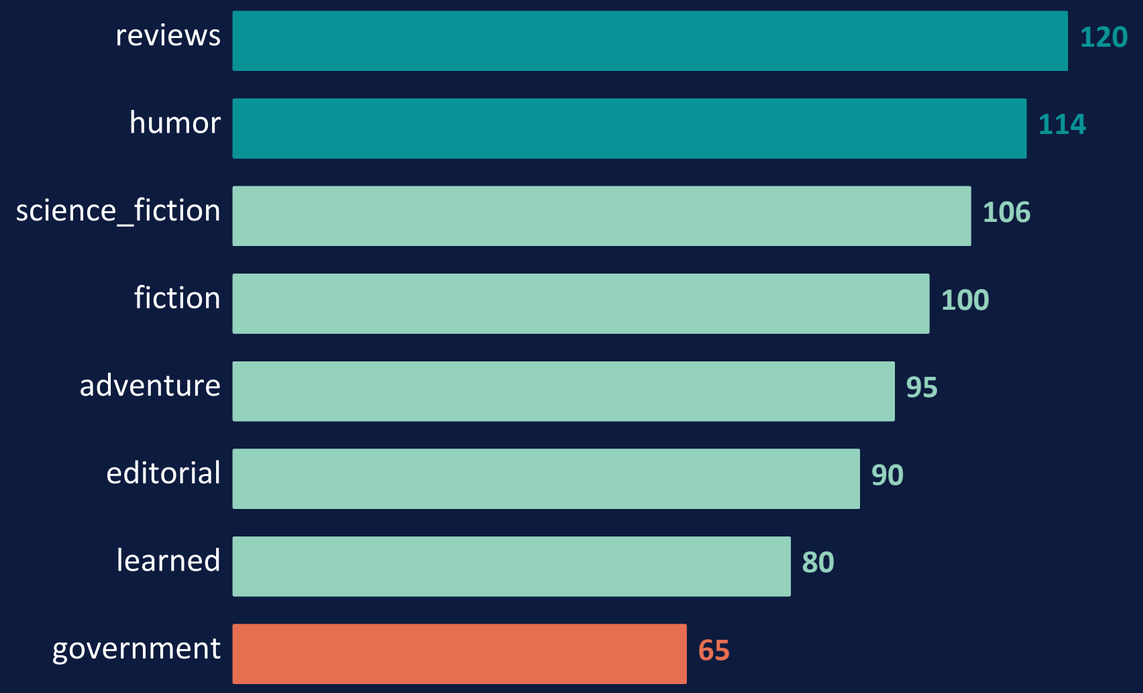
−0.764
TTR–Length Correlation
Proves TTR is length-biased

0.9995
MATTR vs MSTTR (r)
Near-identical signal

0.827
MATTR: Humor (top)
Most lexically diverse genre

~120
MTLD: Reviews (top)
Highest sequential diversity

MTLD Score by Genre (Higher = More Diverse)



Metric Comparison: Top 3 & Bottom Genre

Genre	MATTR	HDD
humor	0.827	0.878
science_fiction	0.822	0.881
reviews	0.820	0.876
...	...	...
government	0.767	0.852

All three robust metrics (MATTR, HDD, MTLD) agree: government is least diverse; reviews/humor/sci-fi are most diverse.

Innovation: Metric Stability Experiment

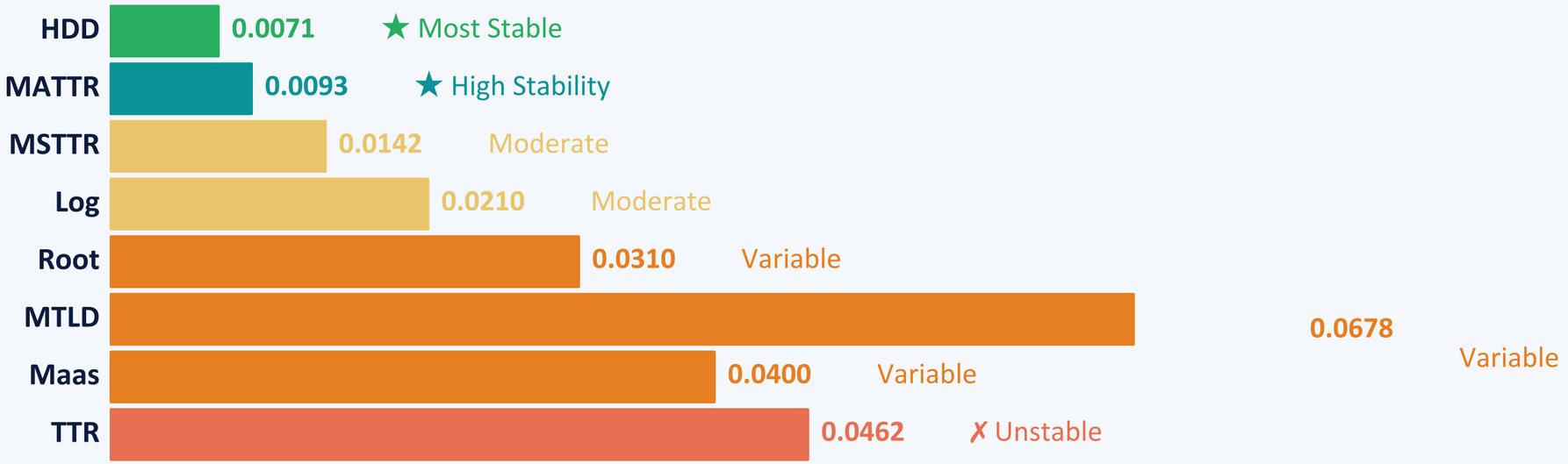
Why is This Innovative?

- Evaluates the metrics themselves, not just the genres
- Tests stability under controlled equal-length sampling
- Provides empirical metric-selection evidence for NLP practitioners
- Goes beyond static corpus comparison — repeated sub-sampling

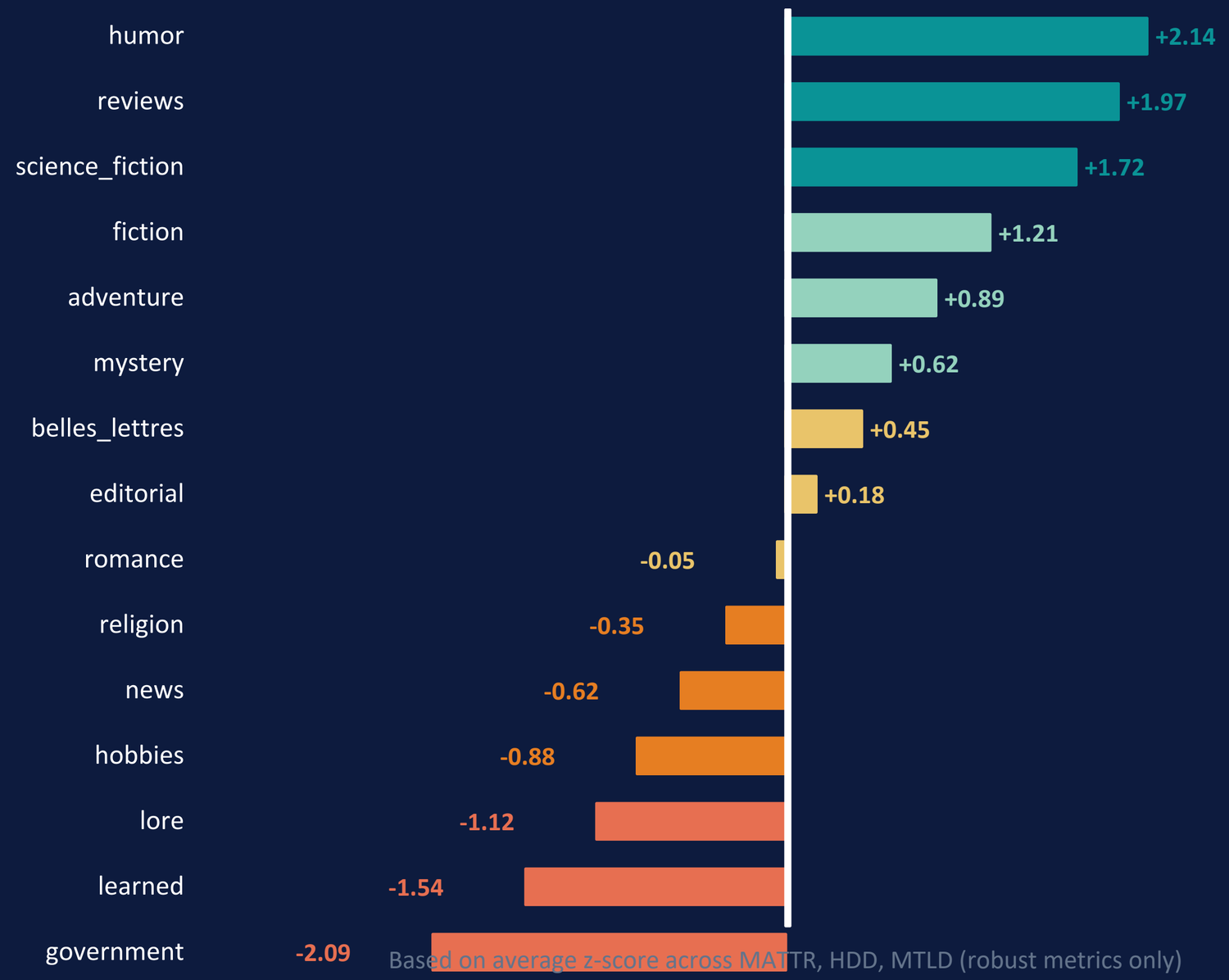
Methodology

1. For each of the 15 genres, repeatedly sample 10,000-token windows (N = 30 iterations)
2. Compute all 8 metrics on each sample
3. Calculate CV = std / mean per metric per genre
4. Lower CV → more stable metric

Stability Results: Mean CV Across 15 Genres



Composite Z-Score Diversity Ranking — All 15 Genres



← *Less Diverse*

Based on average z-score across MATTR, HDD, MTLD (robust metrics only)

More Diverse →

Conclusion

What This Project Show

Lexical diversity varies meaningfully across Brown Corpus genres — humor & reviews are richest; government is most repetitive.

Metric choice is critical. TTR ($r = -0.764$ with length) is unreliable; MATTR, HDD, and MTLD provide fair cross-genre comparison.

Innovation stability experiment confirms: HDD (CV=0.0071) and MATTR (CV=0.0093) are empirically the most stable under repeated sampling.

Composite z-score ranking synthesises all robust metrics into a single interpretable genre ranking — a practical tool for NLP practitioners.

References

1 Francis, W. N., & Kucera, H. (1979). Brown Standard Corpus of Present-Day American English. Providence: Brown University.

2 Covington, M. A., & McFall, J. D. (2010). Cutting the Gordian Knot: The Moving-Average Type–Token Ratio (MATTR). Journal of Quantitative Linguistics, 17(2), 94–100.

3 McCarthy, P. M., & Jarvis, S. (2010). MTLD, vocd-D, and HD-D: A Validation Study of Sophisticated Approaches to Lexical Diversity Assessment. Behavior Research Methods, 42(2), 381–392.

4 Fergadiotis, G., Wright, H. H., & Green, S. B. (2015). Psychometric Evaluation of Lexical Diversity Indices: Assessing Length Effects. Journal of Speech, Language, and Hearing Research, 58(3), 840–852.

5 Jarvis, S. (2013). Capturing the Diversity in Lexical Diversity. Language Learning, 63(1), 87–106.

6 Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media. [NLTK Brown corpus loading]

Dataset also available at: kaggle.com/datasets/nltkdata/brown-corpus